

ALINA ZAMAN

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RESEARCH INTEREST

My research interests lie in **the security and privacy of machine learning systems and software supply chains**. My security interest centers on what happens when proprietary systems depend on external code or data. Professionally, I work on tools for CVE reachability analysis through dependency chains. Currently I am exploring the privacy and robustness challenges of federated learning, including model poisoning attacks on 5G handover prediction. I also have hands-on experience in convolutional feature representation and evaluation through my work on content-based image retrieval.

EDUCATION

Bangladesh University of Engineering and Technology Feb 2020 – March 2025
B.Sc. in Computer Science and Engineering **CGPA : 3.55/4.00 [Senior Year : 3.96/4.00]**

Relevant Coursework: Machine Learning, Computer Security, Computer Networks, Software Engineering, Algorithm Engineering, Basic Graph Theory (**all completed with a 4.00/4.00**)

Shaheed Bir Uttam LT Anwar Girls' College July 2017 – April 2019
Higher Secondary Certificate (HSC) **GPA : 5.00/5.00**

Agargaon Taltola Govt. Colony High School and College Jan 2015 – Feb 2017
Secondary School Certificate (SSC) **GPA : 5.00/5.00**

PUBLICATIONS

[1] Extending Feature Selection Strategies in VGG16: Convolutional Feature Aggregation for Content-Based Image Retrieval, 2026
Saha Kuljit Shantanu, Alina Zaman, Md Monirul Islam
Published at 2026 IEEE 5th International Conference on Computing and Machine Intelligence (ICMI) | [Paper Link](#)

RESEARCH EXPERIENCE

Adversarial Robustness of Federated Learning-Based Handover Prediction

Ongoing Research Project

Supervisors: Assoc. Prof. Sukarna Barua, Dept. of CSE, BUET | Assoc. Prof. Nashid Shahriar, Dept. of CS, University of Regina

- Investigating security vulnerabilities of federated learning models used for handover prediction in 5G networks, including model poisoning, backdoor, and evasion attacks.
- Defining threat models and an experimental framework for evaluating attack impact on network-level metrics.

Attacks and Defenses in Federated Learning

April 2025 – June 2025

Research Collaboration

Mentor: Md Shahedul Haque, then MS student, Virginia Tech

- Following a literature review and paper presentation on attacks and defenses in federated learning, reproduced Bhagoji et al., "Analyzing Federated Learning through an Adversarial Lens" (ICML 2019), on a 10-client federated system built on Adult Census with an MLP implemented from scratch, covering its three model-poisoning attacks (explicit boosting, stealthy poisoning, alternating minimization).
- Across 60 runs (6 attack/defense setups × 5 seeds × 2 target directions), matched the paper's FedAvg results and found that Krum and coordinate-wise median blocked the attacks tested against them in every valid run, unlike the paper's Fashion-MNIST results. [\[Code Link\]](#)

*Undergraduate Thesis**Supervisor: Prof. Md. Monirul Islam, Dept. of CSE, BUET*

- Co-developed an adaptive multi-channel aggregation method for CNN-based image retrieval, replacing the single-dominant-channel selection used by prior work with a dataset-driven threshold that fuses multiple high-ranking convolutional channels.
- Implemented the full retrieval pipeline over frozen VGG16 conv5_3 features: dominance-ranked channel fusion, spatial masking, Gaussian spatial weighting, and PCA-whitening to 100-d descriptors.
- Improved mAP by 2.3 points over the AdCoW+I baseline on both Oxford5K and Paris6K, with gains consistent across all tested threshold values.
- Analyzed the effect of monochromatic input on retrieval, identifying a dataset-dependent effect that motivates follow-up investigation.

PROFESSIONAL EXPERIENCE

OpenRefactory

Software Engineer

Dhaka, Bangladesh

July 2025 – Present, Full-time

- Contributed to the algorithm design and implementation of a VEX generation tool that analyzes CVE reachability using root cause information, dependency chains, and call graphs for both open-source and proprietary code.
- Built a license scanning tool from scratch in Python that detects licenses through static text analysis of license files, copyright headers of source code files, and package metadata, with LLM-based classification as a fallback when static analysis is inconclusive. Output is used to analyze licenses across dependency trees and flag policy violations for open-source license compliance.
- Contributed to the backend and microservices of ProjectClean, including efficient dependency tree generation for artifacts, query optimization for UI performance, and API development.

SKILLS

Programming: Python, C/C++, Java, JavaScript, SQL**Machine Learning:** PyTorch, TensorFlow, Keras, OpenCV**Backend/Web:** Node.js, Spring Boot, React, Next.js**Databases:** PostgreSQL, OracleDB**Tools & Systems:** Git, GitHub, Bash, Linux**Concepts:** OOP, Functional Programming, DevOps

ACADEMIC PROJECTS

Machine Learning Projects | [Code Link](#)

Python, Numpy, Scikit-learn

- **Core ML Algorithms** — Implemented a complete ML pipeline combining data preprocessing, feature engineering, logistic regression with bagging/stacking, PCA, and EM clustering for pattern discovery and model evaluation.
- **Multi-Layer Perceptron (MLP)** — Implemented a feed-forward neural network from scratch, incorporating dense layers, batch normalization, dropout regularization, and the Adam optimizer to improve convergence and generalization during training.

Computer Security & Cybersecurity Projects | [Code Link](#)

Python, Docker, Wireshark, Azure

- **Cryptography & Key Exchange** — Implemented AES encryption/decryption and elliptic-curve key generation in Python; simulated secure socket-based communication.

- **Web Security & Browser Exploitation** — In a controlled lab environment (SEED Labs), demonstrated CSRF and XSS attacks against a local test application.
- **Malware Signature Analysis with YARA** — Created and organized YARA rules for ELF, PE, console, and test-based detection patterns for static malware analysis and signature matching.

Computer Networking Projects | [Code Link](#)

Java, Cisco Packet Tracer, NS3

- **Client-Server Networking & Reliable Data Transfer** — Implemented Java socket-based applications supporting multi-client communication, threaded servers, FTP-style file transfer, and reliable data transmission using CRC and Hamming code.
- **Network Simulation with ns-3** — Implemented an ns-3 based wireless network simulation to evaluate TCP performance under varying traffic, topology, and mobility conditions through throughput and packet delivery analysis

Computer Graphics Projects | [Code Link](#)

Graphics, OpenGL

- **OpenGL Rendering & Animation** — Built 3D scenes with primitives, transformations, camera setup, and animated motion, including rolling/board-like scenes and cube/cylindrical object constructions.
- **3D Transformation Pipeline** — Implemented translation, rotation, scaling, and projection transformations; processed triangle data through multiple rendering stages and performed z-buffer visibility checks.
- **Ray Tracing & Shading** — Built a renderer that shoots rays from the camera through pixels, intersects them with spheres and triangles, applies ambient/diffuse/specular lighting, and exports the final images as BMP files.

C Compiler | [Code Link](#)

Flex, Bison, YACC, C++

- Built a mini-compiler for a C-like language, covering lexical analysis, parsing, symbol table and scope management, and basic assembly-style code generation for expressions, control flow, and functions.

Thoughts | [Code\(Frontend\)](#) [Code\(Backend\)](#)

React.js, Spring Boot

- Developed a web application integrating Spring Boot, STOMP, and LLaMA 3.2 for real-time communication and LLM-powered assistance, inspired by platforms like *The Help Line*.

OTHER EXPERIENCES

Bangladeshi Women in Computer Science and Engineering (BWCSE)

2023 – 2025

Served as the *Batch Representative* for two consecutive years.

Student Tutor

2020 – 2025

Provided academic tutoring and mentorship to students from middle school through undergraduate level, including guidance and preparation for university admissions.

REFERENCES

Munawar Hafiz, CEO at OpenRefactory, Inc.

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Dr. Md. Monirul Islam, Professor at BUET

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Sukarna Barua, Associate Professor at BUET

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